Inverse Problems and Imaging - 02624

Homework 3

1. The 2D heat problem on $(0,\pi)^2$ with homogeneous Dirichlet boundary conditions has the forward operator

$$Kx(t_1,t_2) = \int_0^\pi \int_0^\pi k(t_1,t_2,s_1,s_2)x(s_1,s_2)\mathrm{d}s_1\mathrm{d}s_2$$

with kernel

$$k(t_1,t_2,s_1,s_2) = \sum_{n,m} e^{-(n^2+m^2)T}\sin(ns_1)\sin(ms_2)\sin(nt_1)\sin(mt_2).$$

It takes as input the image $x(s_1,s_2)$ as initial condition and outputs the heat distribution at time T, $y(t_1,t_2) = Kx(t_1,t_2)$.

   (a) Describe in an essay text (max 5 lines) why the 2D backwards heat equation is ill-posed.

   (b) Implement numerically uing FFT a numerical approximation A of the operator K as a function (in Python/Matlab) that takes as input an N by N matrix x of sampled values of x and returns an N by N matrix y approximating $y(t_1,t_2)$. Show that the function acts correct on $x(s_1,s_2) = \sin(ks_1)\sin(ls_2)$.

   (c) Download the image `KarinHarbo512.png` of size N^2, $N = 512$. Use this as x. Fix $T = 0.001$ and compute $y = Kx$. Add random Gaussian noise to y of magnitude $\delta = 0.01$ to find y^δ. Display y^δ.

   (d) Show that K is selfadjoint, and hence use $K^* = K = A$. Solve the inverse problem using the Landweber iteration (take 1500 iterations); look at the residual error history.

2. Revisit exercises 1-3 from the program in week 12.

   (a) Download the image `KarinHarboStripe.png` and import to Python/Matlab as matrix of size 100x100 and plot it.

   (b) Fill the black rectangular region by TV inpainting and harmonic impainting. Desribe birefly the two methods.

   (c) Plot the results, compare and explain.

Important notes:

- Deadline for the HW3 is Friday December 5.
- All homeworks must be approved in order to qualify for the exam.

- Your background is different, and therefore you can choose to put emphasiz differently in your solution.
- You must demonstrate your understanding of the material.
- Each student should hand in a single pdf via Learn. The number of pages for the main report must not exceed 7. Code can be put in an appendix.
- Comment on all results and plots: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions.